

Thompson Rivers University

Forecasting Mean Temperature Using LSTM and
ARIMA: An Analysis of Australian Weather Data

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1 Abstract

This study evaluates the forecasting performance of ARIMA and LSTM models using the Temperature Rain Dataset with Missing Values, comprising 32,072 daily time series collected from 422 weather stations across Australia. After rigorous preprocessing, including data imputation and normalization, a subset of 100 time series was selected to ensure computational efficiency. The ARIMA model, known for its strength in modeling linear relationships, demonstrated robust performance in capturing stationary trends and seasonal patterns. In contrast, the LSTM model excelled in handling non-linear dependencies and long-term temporal relationships, leveraging its ability to process sequences with missing values. The evaluation, conducted across 1-day, 3-day, and 7-day forecast horizons, revealed that ARIMA achieved lower Mean Squared Error (MSE) for short-term forecasts, while LSTM outperformed in capturing complex patterns over longer periods. The results underscore the complementary nature of statistical and deep learning approaches in time series forecasting. This work highlights the importance of model selection based on the underlying data characteristics and forecasting objectives, providing insights for future research on hybrid modeling strategies to enhance predictive accuracy in meteorological datasets. The code and additional resources related to this report can be found on the following GitHub repository created by Yueyue Zhang, Sepideh Mansourigovari, Begimai Alisherova, 2024.

2 Introduction

Time series forecasting is a critical task in meteorology, providing insights for decision-making in agriculture, disaster management, and urban planning. The Temperature Rain Dataset with Missing Values offers a unique opportunity to

explore advanced forecasting techniques by leveraging real-world data with temporal dependencies and missing entries. This dataset, comprising temperature observations from 422 weather stations across Australia, represents a challenging and practical testbed for developing and evaluating forecasting models.

In this study, our aim is to implement and compare two widely used forecasting methods: the Auto-Regressive Integrated Moving Average (ARIMA) model and Long Short-Term Memory (LSTM) networks. ARIMA, a traditional statistical model, has been widely used for its simplicity and interpretability in linear time series prediction (Box et al., 2015). On the other hand, LSTMs, a class of recurrent neural networks, excel in capturing long-term dependencies and nonlinear patterns in complex datasets (Hochreiter, Schmidhuber, 1997). By applying these methods to the dataset, this research seeks to address the challenges of missing value and uncover patterns in temperature trends, providing a reference point for further studies.

3 Data

The dataset used in this study is the *Temperature Rain Dataset with Missing Values* taken from Australian Bureau of Meteorology, 2017, a comprehensive collection of meteorological data designed to support research in time series forecasting and data imputation. The dataset contains 32,072 daily time series, including temperature observations and rainfall forecasts, recorded across 422 weather stations in Australia. The data spans from May 2, 2015, to April 26, 2017, obtained from the Australian Bureau of Meteorology.

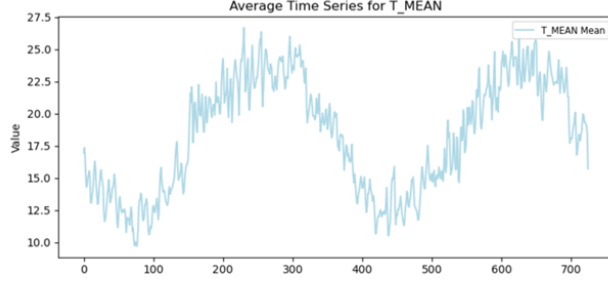


Figure 1: Average Time Series for T_MEAN

In this study, a random subset of 100 time series was selected to ensure computational efficiency while retaining diverse characteristics of the dataset. These time series represent temperature observations (T_MEAN) from various weather stations, each with daily measurements. Given the prevalence of missing values in the dataset, several preprocessing steps were applied to prepare the data for modeling:

- (1) **Data Cleaning:** The raw data contained missing values represented by placeholder symbols (e.g., “?”). These missing values were replaced with NaN to facilitate subsequent handling.
- (2) **Time Series Continuity:** Each selected time series was checked to ensure continuity of timestamps. Missing timestamps were identified and filled to reconstruct complete daily sequences.
- (3) **Data Imputation:** Missing values within the time series were imputed using linear interpolation, ensuring smooth transitions in the data while preserving temporal trends.
- (4) **Normalization:** To standardize the data for machine learning models, all time series values were scaled to a range between 0 and 1 using Min-Max Scaling.

- (5) **Sequence Preparation:** For models requiring sequential input, such as LSTM, overlapping subsequences of a fixed length (e.g., 28 days) were created. These subsequences served as the input for predicting future observations.

Through these preprocessing steps, the dataset was transformed into a format suitable for time series modeling, allowing for the evaluation of forecasting algorithms under realistic conditions of missing data and temporal dependencies.

4 Methodology

4.1 Auto-Regressive Integrated Moving Average (ARIMA)

The Auto-Regressive Integrated Moving Average (ARIMA) model was chosen for its effectiveness in modeling linear relationships within time series data by leveraging temporal dependencies. For the Temperature Rain Dataset with Missing Values, ARIMA was applied to univariate time series data, such as daily temperature observations. The methodology included the following steps:

(1) Data Preparation:

- Missing values were imputed using linear interpolation to ensure data continuity.
- Each time series was transformed into a stationary form using differencing to avoid inaccuracies caused by non-stationary data. The order of differencing (d) was determined by analyzing the autocorrelation function (ACF) and partial autocorrelation function (PACF).

(2) Model Selection:

- Model parameters were selected using the `auto_arima` function from the `pmdarima` library, optimizing the following parameters:
 - p : The order of the auto-regressive (AR) term.
 - d : The differencing order for stationarity.
 - q : The order of the moving average (MA) term.
- Seasonal components were disabled (`seasonal=False`) as the dataset lacked evident seasonal patterns and spanned a relatively short time frame.

(3) Validation and Forecasting:

- The ARIMA model was trained on sequences of up to 28 days prior to the forecast horizon, with forecasts made for 1-day, 3-day, and 7-day windows.
- The model’s performance was assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE).

4.2 Long Short-Term Memory (LSTM) Networks

Long Short-Term Memory (LSTM) networks, a type of recurrent neural network, were employed to capture the non-linear dynamics and long-term dependencies inherent in the dataset. The methodology comprised the following steps:

(1) Data Preparation:

- Missing values were imputed using linear interpolation.
- Each time series was normalized to a range of 0 to 1 using Min-Max Scaling to enhance model convergence during training.
- Sliding windows of 28-day sequences were created as inputs for the LSTM model, with the corresponding next values as target outputs.

(2) Model Architecture:

- The LSTM model consisted of two stacked LSTM layers, each with 50 hidden units, followed by a dense output layer with a single neuron for regression:

```
model = Sequential()
model.add(LSTM(units=50, return_sequences=True, input_shape=(28, 1)))
model.add(LSTM(units=50))
model.add(Dense(1))
```

- The Adam optimizer was used to manage sparse gradients and adaptive learning rates, with the Mean Squared Error (MSE) as the loss function.

(3) Hyperparameter Tuning:

- Training was capped at 50 epochs with early stopping to prevent overfitting, using a patience of 5 epochs based on validation loss.
- Batch size was set to 32 to balance memory usage and training efficiency.

(4) Validation and Forecasting:

- The model was trained on sequences of up to 28 days prior to the forecast horizon, making predictions for 1-day, 3-day, and 7-day windows.
- The model's performance was evaluated using MSE, MAE, and MAPE metrics.

Through these methodologies, the study assesses the strengths and limitations of ARIMA and LSTM models in handling incomplete time series data and diverse temporal dependencies, offering insights into their applicability for weather-related forecasting tasks.

5 Results

The comparison between ARIMA and LSTM models in forecasting daily temperature observations is presented for three forecast horizons: 1-day, 3-day, and 7-day. The results, visualized in Figures 1 through 6, provide insights into the performance and suitability of these models.

For the 1-day forecast horizon, ARIMA demonstrates superior performance over LSTM in terms of Mean Squared Error (MSE) and Mean Absolute Error (MAE). However, LSTM exhibits a slightly higher Mean Absolute Percentage Error (MAPE), indicating its relative struggle with percentage-based deviations for short-term predictions.

For the 3-day horizon, ARIMA continues to outperform LSTM in MSE and MAE. Nevertheless, LSTM achieves a slightly lower MAPE compared to ARIMA, suggesting better adaptability to relative variations in mid-term forecasts.

For the 7-day forecast horizon, the errors for both models increase, reflecting the challenges associated with long-term forecasting. ARIMA maintains a lower MSE, while LSTM exhibits marginally better MAPE, highlighting its ability to capture non-linear relative changes despite higher absolute deviations.

The error trends across forecast horizons reveal distinct characteristics of the two models. LSTM shows a more pronounced increase in MSE, MAE, and MAPE as the forecast horizon extends, reflecting its sensitivity to temporal dependencies and challenges in maintaining stability for long-term predictions. Conversely, ARIMA demonstrates a steadier growth in error values, underscoring its robustness in capturing linear trends across varying forecast horizons.

The findings highlight the complementary strengths of ARIMA and LSTM models. ARIMA is well-suited for short-term forecasts, where linear trends dominate, and its lower computational requirements make it a practical choice

for simpler datasets. On the other hand, LSTM's ability to model non-linear dynamics provides an advantage in datasets with irregular trends and patterns, though its performance in long-term forecasting requires further optimization.

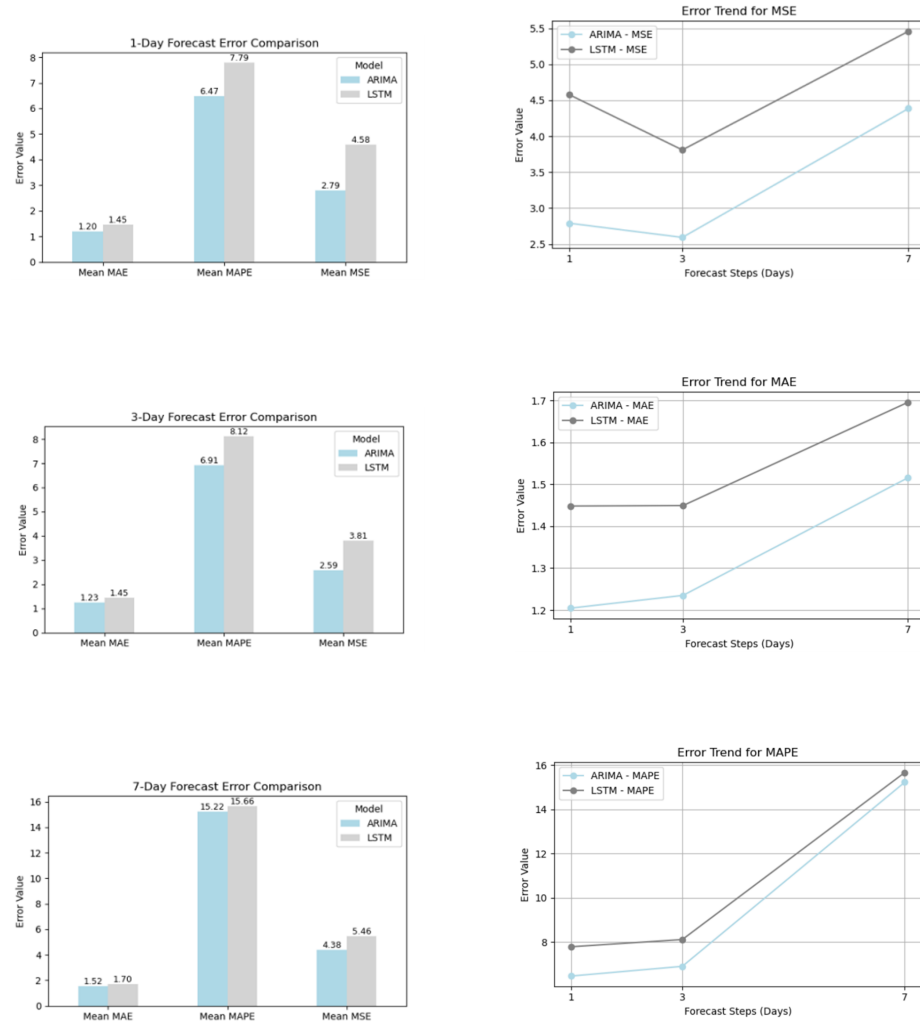


Figure 2: Comparison of Forecast Errors and Trends for ARIMA and LSTM Models

6 Discussions and Conclusions

In this study, we compared the performance of ARIMA and LSTM models for forecasting daily temperature observations from the Temperature Rain Dataset with Missing Values. ARIMA demonstrated robust performance with lower MSE and MAE across all forecast horizons, particularly excelling in capturing linear trends and short-term dependencies. LSTM, on the other hand, effectively modeled non-linear patterns and long-term dependencies but showed higher absolute errors in extended forecasts, highlighting its sensitivity to accumulating temporal complexities. The findings suggest that ARIMA is more suitable for datasets with consistent patterns and short-term forecasting needs, while LSTM is better suited for complex, non-linear datasets despite requiring further optimization for longer-term predictions. Future research could explore hybrid approaches to leverage the strengths of both models.

References

- Australian Bureau of Meteorology. (2017). Temperature rain dataset with missing values [Accessed: 2024-12-01]. <https://zenodo.org/record/5129073>
- Yueyue Zhang, Sepideh Mansourigovari, Begimai Alisherova. (2024). GitHub Repository for Phishing URL Detection: Codes and Results [Accessed: 2024-12-01]. <https://github.com/begimai-alisherova/Time-Series-Project.git>